

EBN111 CT2 Memo

Sunday, 08 May 2022 13:37

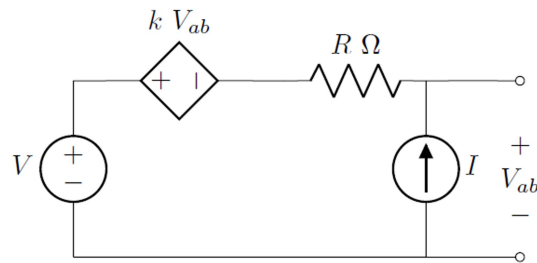
Questions 1 to 3 [4]

Given: $R = 15 \Omega$, $V = 6 \text{ V}$, $I = 1 \text{ A}$ and $k = 4$.

Use superposition to determine:

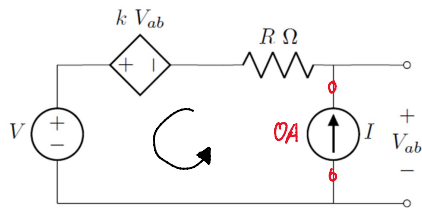
1. the contribution of the independent voltage source to the V_{abV} exclusively.
2. the contribution of the independent current source to V_{abI} exclusively, and

Determine the total V_{ab} as the result of both the current and voltage source.



01	Independent contribution from Voltage source V: V_{abV} [1]	1.2 V
02	Independent contribution from Current source I: V_{abI} [2]	3 V
03	Total contribution to: V_{ab} [1]	4.2 V

1



V_{abV} : KVL

$$V - v_{abV} + R(i) - k v_{abV} = 0 \quad (1)$$

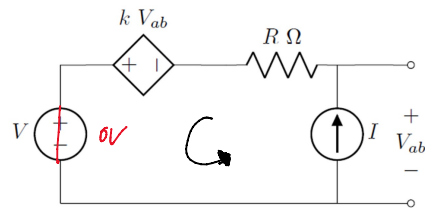
$$i = 0$$

$$\therefore V = v_{abV}(1+k)$$

$$\Rightarrow v_{abV} = \frac{V}{1+k} \text{ V}$$

$$= \frac{6}{1+4} = 1.2 \text{ V}$$

2



$$V_{abI}: -v_{ab} + R(i) - k v_{ab} = 0$$

$$i = I$$

$$\Rightarrow RI = v_{ab}(1+k)$$

$$\Rightarrow v_{abI} = \frac{RI}{1+k}$$

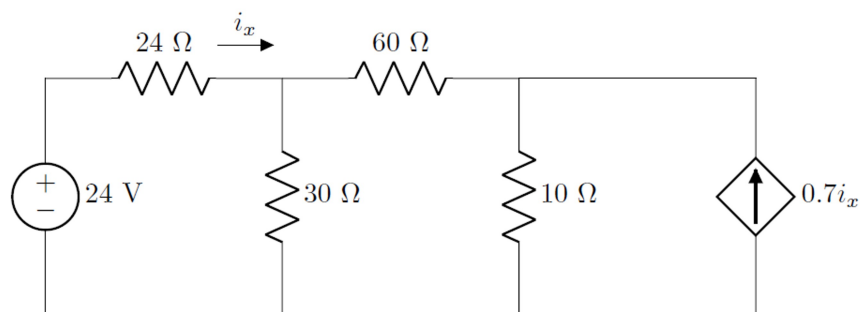
$$= \frac{15 \cdot 1}{1+4} = 3 \text{ V}$$

3

$$V_{ab} = V_{abV} + V_{abI} = 1.2 + 3 = 4.2 \text{ V}$$

Question 4 [2]

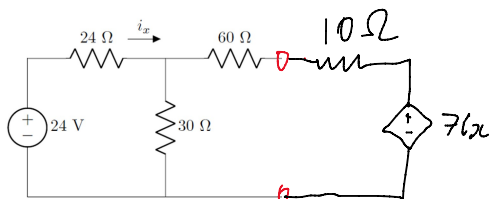
Use source transformation to determine the current of i_x .



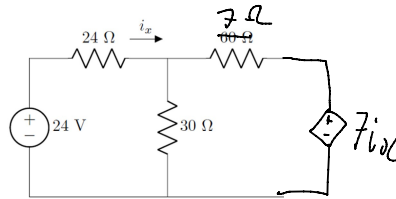
04

i_x [2]

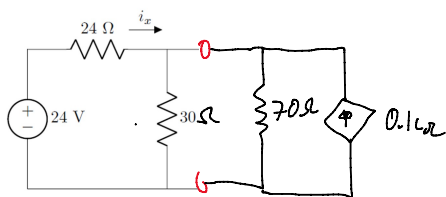
509.55mA



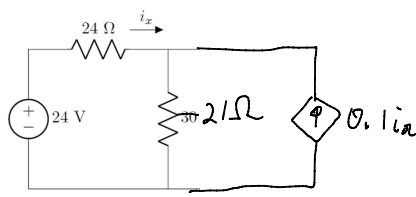
$$V = iR = 7i_x V$$



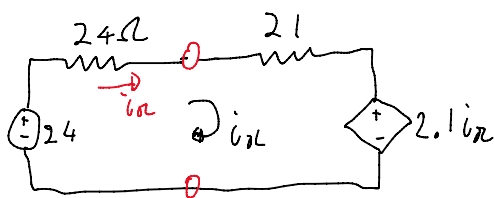
$$R_{eq} = 70\Omega$$



$$i = V/R = \frac{24}{70} i_x = 0.1i_x$$



$$R_{eq} = 70 || 30 = 21$$



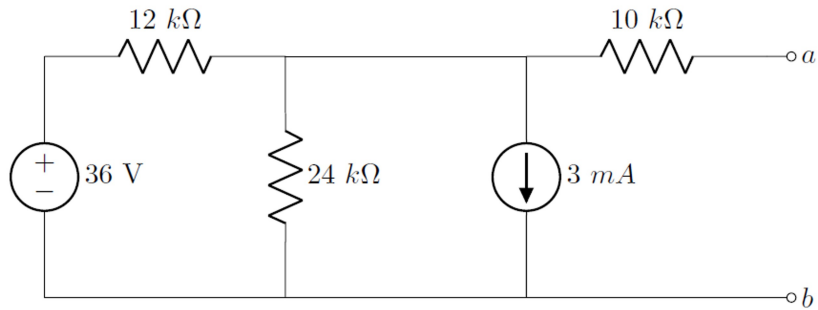
$$-24 + 24i_x + 21i_x + 2.1i_x = 0$$

$$24 = 47.1 i_x$$

$$\therefore i_x = \frac{24}{47.1} = 509.55 \text{ mA}$$

Questions 5 to 6 [4]

Use Norton's Theorem to determine the Norton equivalent circuit with respect to terminals a and b.



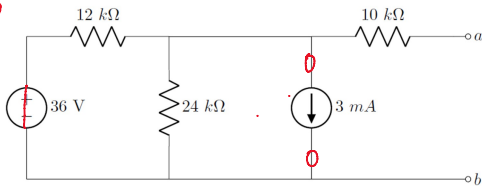
05 R_N [2]

$18k\Omega$

06 I_N [2]

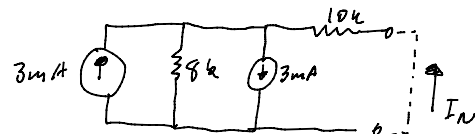
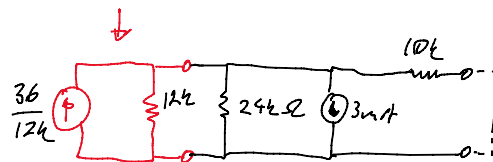
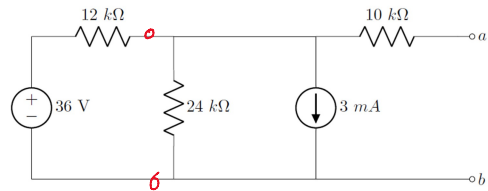
$0A$

5



$$\begin{aligned} R_{TH} &= (12k \parallel 24k) + 10k = \\ &= 8k + 10k \\ &= 18k \end{aligned}$$

6

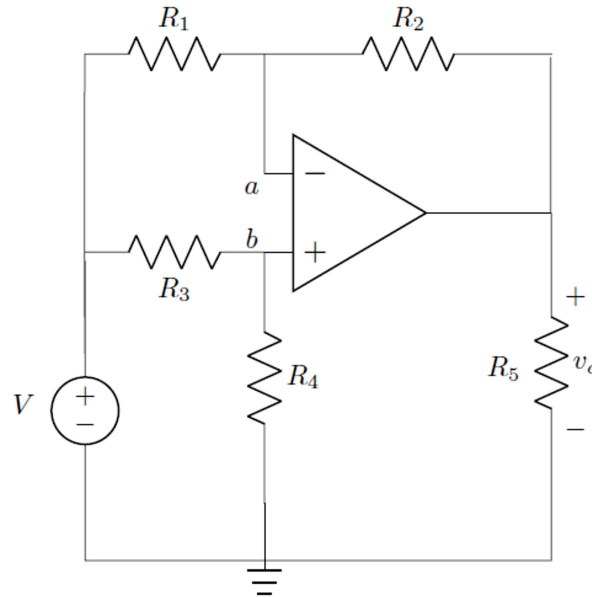


$$I_N = 0A = I_1 + I_2$$

Questions 7 to 8 [3]

Given: $R_1 = 2k\ \Omega$, $R_2 = 8k\ \Omega$, $R_3 = 5k\ \Omega$, $R_4 = 10k\ \Omega$, $R_5 = 2k\ \Omega$ and $V = 3\text{ V}$.

Determine the output voltage (v_o) and the voltage at node b (V_b) of the amplifier.



07 Determine V_b . [1] 2V

08 Determine v_o . [2] -2V

$$\begin{aligned}
 7 \quad V_b &= \frac{R_4}{R_4 + R_3} \cdot V \\
 &= \frac{10k}{10k + 5k} \cdot 3 \\
 &= \frac{2}{3} \cdot 3 = 2V
 \end{aligned}$$

$$V_a = V_b = 2V \rightarrow$$

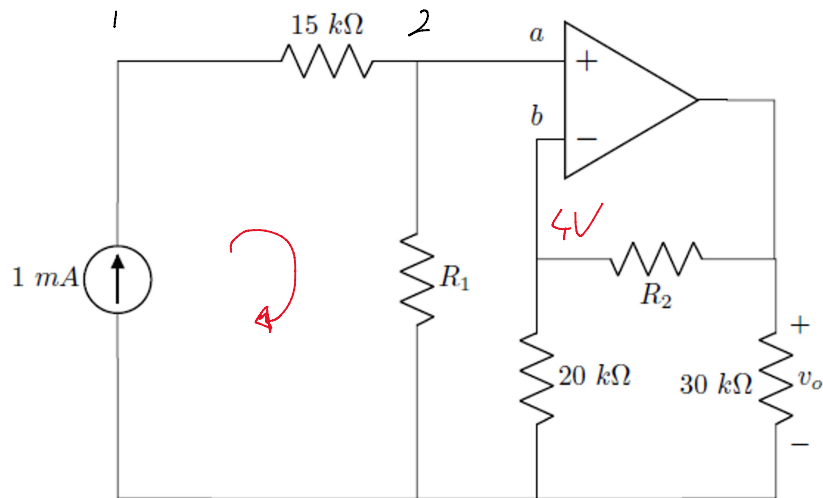
8 Diff op amp

$$\begin{aligned}
 \frac{V - V_a}{R_1} &= \frac{V_a - V_o}{R_2} \\
 \Rightarrow V_a - V_o &= \frac{R_2}{R_1} (V - V_a) \\
 \Rightarrow V_o &= -\frac{R_2}{R_1} (V - V_a) + V_a \\
 V_o &= -\frac{8k}{2k} (3 - 2) + 2 \\
 &= -4(1) + 2 \\
 &= -2V \rightarrow
 \end{aligned}$$

Questions 9 to 10 [3]

Given: $R_1 = 4 \text{ k}\Omega$ and $R_2 = 10 \text{ k}\Omega$

Determine the voltage at the non-inverting input (v_a) of the op-amp and the output voltage (v_o) of the op-amp circuit shown in the circuit diagram below.



09 V_a [1]

4V

10 V_o [2]

6V

9

$$V_a = 4 \text{ k} \times 1 \text{ mA} = 4 \text{ V}$$

$$1 \text{ mA} = \frac{v_1 - v_2}{15 \text{ k}}$$

$$0 = \frac{v_2 - v_1}{15 \text{ k}} + \frac{v_2 - 0}{4 \text{ k}}$$

$$\Rightarrow -1 \text{ mA} + \frac{v_2}{4 \text{ k}} = 0$$

$$v_2 = 1 \text{ mA} \times 4 \text{ k} = 4 \text{ V}$$

$$v_2 = v_a \therefore \underline{v_a = 4 \text{ V}}$$

10

$$v_a = v_b = 4 \text{ V}$$

$$\frac{v_o}{30 \text{ k}} + \frac{v_o - v_b}{10 \text{ k}} = 0$$

$$\frac{v_b - v_o}{10 \text{ k}} + \frac{v_b - 0}{20 \text{ k}} = 0$$

$$\Rightarrow 2v_b - 2v_o + v_b = 0$$

$$\Rightarrow 2v_o = 3v_b$$

$$\Rightarrow v_o = \frac{3}{2} v_b = \frac{3}{2} (4) = \underline{6 \text{ V}}$$